Maham Khan

Senior Data Engineer | Cloud-Native Data Platforms | GenAI & ML Infrastructure

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Professional Summary

Senior Data Engineer with 10+ years of experience architecting TB-to-PB-scale, cloud-native data platforms for enterprise environments in Healthcare and Technology (NVIDIA). Proven record of delivering measurable business outcomes: 3x query performance gains, 40% fewer pipeline failures, 50% faster time-to-insight, and $1.2M+ in annual infrastructure savings. Production architect for GenAI data infrastructure - RAG pipelines, vector search (Pinecone, Weaviate), embedding generation, and feature stores supporting 5+ models across 10+ business units. Deep expertise in HIPAA/PHI, GDPR, and SOX-compliant environments.

Core Technical Skills

Languages and Query: Python, SQL, PySpark, Scala, Java, Bash

Cloud Platforms: Azure (ADF, ADLS Gen2, Synapse, Azure DevOps) **|** AWS (S3, Glue, Redshift, EMR, Kinesis, Lambda, SageMaker) **|** GCP (BigQuery, Dataflow, Composer, Dataproc)

Big Data and Streaming: Apache Spark, PySpark, Databricks, Delta Lake, Delta Live Tables, Apache Kafka, Spark Structured Streaming, Apache Flink, AWS Kinesis, Azure Event Hubs

Data Warehouses and Stores: Snowflake, Databricks Lakehouse, BigQuery, Redshift, Azure Synapse, Apache Iceberg, PostgreSQL, Teradata, Oracle

ETL and Orchestration: Apache Airflow, dbt Core and Cloud, Azure Data Factory, AWS Glue, Fivetran, Airbyte, Informatica, Talend, SSIS

Architecture Patterns: Medallion Architecture (Bronze/Silver/Gold), Data Lakehouse, Data Mesh, Dimensional Modeling, Star/Snowflake Schema, Data Vault 2.0, Semantic Layer, MDM

GenAI and ML Infrastructure: RAG Pipelines, Embedding Generation, Vector Databases (Pinecone, Weaviate, Chroma), Feature Stores, MLflow, LLM Data Pipelines, AI Observability

Data Governance and Security: Unity Catalog, RBAC, Data Lineage, PII/PHI Controls, Data Quality Frameworks, HIPAA, GDPR, SOX, Audit Controls

DevOps and CI/CD: Git, GitHub Actions, Azure DevOps, Terraform, Docker, Kubernetes, Linux

BI and Analytics: Power BI, Tableau, Looker / LookML, Amazon QuickSight

Integrations: SAP S/4HANA, SAP BW, Salesforce, HubSpot, NetSuite, REST APIs, Microservices

Professional Experience

Senior Data Engineer | NVIDIA

Remote **|** *Santa Clara, CA* **|** *Jan 2024 - Present*

- Architected a production GenAI data platform - RAG pipelines, embedding generation, and vector database integration (Pinecone, Weaviate) - supporting 5+ AI models across 10+ business units; reduced model inference retrieval latency by 45% and cut embedding pipeline processing time by 35%.
- Governed Databricks and Snowflake lakehouses end-to-end using medallion architecture (Bronze/Silver/Gold) with Delta Live Tables and Unity Catalog, processing 500+ TB/month of structured, semi-structured, and unstructured data with full lineage, RBAC, and PII/PHI controls.
- Standardized dimensional modeling frameworks (star/snowflake schema, semantic layer, Data Vault 2.0) across 10+ business units - improving average query performance 2-3x and reducing ad hoc requests to the engineering team by 60%.
- Designed ingestion pipelines from SAP S/4HANA, SAP BW, Salesforce, and 10+ external platforms processing 50M+ records per day with full audit trails; reduced downstream reconciliation errors by 35% through automated validation and lineage controls.

- Led on-premise-to-cloud migration consolidating 4 legacy systems onto Azure, Databricks, and Snowflake - reducing infrastructure costs by 32% (~$1.2M annually) while improving scalability for GPU-accelerated AI workloads.
- Built semantic layers and self-service analytics platforms enabling 10+ teams to access KPI reporting independently - cutting average time-to-insight from 3 days to under 4 hours and saving analysts 15+ hours per week.
- Established data quality frameworks with 200+ automated validation rules and real-time alerting - reducing data incidents reaching downstream consumers by 55% within the first quarter of deployment.
- Mentored 6 engineers on architecture standards, governance frameworks, and GenAI data patterns; authored platform documentation adopted as the internal standard across 3 engineering teams.

Senior Data Engineer | Deaconess Health System

Evansville, IN | *Remote* | *Jan 2019 - Dec 2023*

- Engineered HIPAA-compliant data platforms across Azure, AWS, Snowflake, and BigQuery - processing 200+ TB of clinical and operational data annually with PHI masking, RBAC, and full audit trails across EHR, claims, and operational sources.
- Implemented medallion architecture and dbt-based dimensional models across 3 clinical domains - improving query performance 3x and enabling self-service analytics for 8 teams in clinical operations, revenue cycle, and population health.
- Built Kafka and Spark Structured Streaming pipelines processing 5M+ healthcare events per day - reducing data latency from 6 hours to under 10 minutes and enabling near-real-time clinical reporting for 4 critical care dashboards.
- Implemented data quality and monitoring frameworks with automated validation and alerting across 30+ production pipelines - reducing failures by 40% and maintaining 99.9%+ SLA adherence over a 4-year period.
- Developed AI/ML feature engineering pipelines and model training infrastructure for 3 clinical decision support models - reducing feature preparation time by 50% and supporting inference pipelines serving 500+ clinicians daily.
- Tuned Apache Spark workloads via partition pruning, broadcast joins, and caching - improving execution performance 30-50% and reducing monthly cloud compute spend by ~$180K (25%) through resource right-sizing.
- Built CI/CD pipelines using GitHub Actions and Azure DevOps - reducing deployment time from 4 hours to under 45 minutes and cutting production incidents by 35%.
- Mentored 4 junior engineers and drove adoption of dbt, Airflow, and Fivetran - increasing team throughput by 30% and cutting new-engineer onboarding time from 6 weeks to 3 weeks.

Data Engineer | RevSpring

Newtown Square, PA | *Jan 2016 - Dec 2018*

- Built ETL/ELT pipelines in Python, PySpark, and SQL processing 100+ TB across Redshift, BigQuery, and Snowflake - incremental load patterns and metadata-driven frameworks reduced full-load runtimes by 65% and cut processing costs by 28%.
- Developed dbt-based dimensional models across 5 reporting domains - improving query performance 3x and reducing BI team report delivery time from 2 weeks to 3 days.
- Implemented Kafka event streams processing 1M+ events per day with end-to-end latency under 30 seconds - enabling real-time operational analytics for time-sensitive business use cases.
- Optimized Spark jobs via partition pruning, broadcast joins, and execution plan tuning - reducing average job runtime 30-40% and saving ~$90K annually in cloud compute costs.
- Delivered optimized datasets for Tableau and Power BI dashboards serving 200+ business users - improving dashboard load time by 55% and reducing ad hoc warehouse query volume by 40%.

Associate Data Engineer | Devsinc

Remote | *Jan 2014 - Dec 2015*

- Built ETL/ELT pipelines integrating Salesforce, HubSpot, REST APIs, and 6+ enterprise data sources across AWS and Azure - reducing manual data preparation effort by 70% and supporting 3 product lines.
- Developed dimensional models and transformations supporting BI dashboards and operational reporting - cut recurring data preparation time by 70% through pipeline automation.
- Implemented data lineage tracking and metadata management, improving audit request response time by 50%.

Projects

Enterprise Lakehouse Modernization **- Azure | Databricks | Snowflake | Unity Catalog | Delta Live Tables**

Consolidated 4 legacy on-premise systems into a medallion lakehouse processing 500+ TB/month from SAP S/4HANA, Salesforce, and 10+ external platforms. Deployed Unity Catalog for enterprise-wide governance and RBAC. Outcomes: 32% cost reduction (~$1.2M/year), 45% lower data latency, 3x query performance improvement, and self-service analytics scaled across 10+ business domains.

Real-Time Streaming and Data Quality Platform **- Apache Kafka | Spark Structured Streaming | AWS | GCP**

Designed and operated streaming pipelines processing 5M+ events/day with end-to-end latency under 30 seconds. Deployed 200+ automated validation rules that reduced downstream data incidents by 55% and enabled near-real-time analytics across 4 critical operational dashboards.

GenAI and LLM Data Platform **- Feature Store | RAG Pipelines | Pinecone | Weaviate | MLflow | Databricks**

Built AI/ML data infrastructure supporting 5+ production models - RAG pipelines, embedding generation, vector database integration, and semantic search at scale. Reduced inference retrieval latency by 45% and embedding pipeline runtime by 35%. Implemented full MLflow observability and model lineage across all workloads.

HIPAA-Compliant Clinical Data Platform **- Azure | Snowflake | dbt | Apache Airflow | Kafka**

Processed 200+ TB/year of EHR, claims, and operational data with PHI masking, RBAC, and full audit controls. Reduced data latency from 6 hours to under 10 minutes, maintained 99.9%+ SLA adherence, and cut pipeline failures by 40% across 30+ production pipelines over a 4-year period.

Certifications and Professional Development

- Databricks Certified Associate Developer for Apache Spark - Completed
- dbt Analytics Engineering - Advanced coursework completed
- Apache Kafka Fundamentals - Completed
- MLflow for ML Lifecycle - Completed
- AWS Certified Data Analytics - Specialty (in progress, 2026)
- Microsoft Certified: Azure Data Engineer Associate DP-203 (in progress, 2026)

Education

Bachelor of Science, Computer Engineering | Manhattan University | 2013